# Virtual Memory

The Kiwi™ memory architecture design team has a dilemma. The team is considering several different memory configuration variations for an upcoming machine design. Consider the following designs (All memory accesses are in terms of bytes, and all are using paging techniques):

| **Characteristic** | **Design 1** | **Design 2** | **Design 3** |
|---|---|---|---|
| Physical Memory Address Width (#bits in physical address) | 10 bit | 20 bit | 24 bit |
| Virtual(Logical) Address Width (#bits in virtual address) | 16 bit | 32 bit | 36 bit |
| Page size in bytes | 32 bytes | 64 bytes | 128 bytes |
| Page Table Type | Single | Single | Double |

(Single: only one page table. Double: 2-level page table. )

a) For each design, list the maximum number of pages each process can access in logical address space. (hint: How many possible virtual pages? Logical address means virtual address.)

b) For each design, list the maximum number of pages in physical memory. (hint: How many possible physical pages?)

c) For design 3, if the outermost page table holds 128 entries, how many bits are needed in the logical address to represent the outer page table? How many bits are used for representing the offset within a page? How many bits are needed in the logical address in order to represent the inner page table?